DE–5830 **11**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2013.

ELECTRONIC DEVICES AND DIGITAL CIRCUIT

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What is the full form of ASCII? What is the maximum number of characters, which can be representated by these codes? Discuss. (5)

   (b) Draw the logic diagram of arithmetic circuit and explain. (6)

   (c) Convert the following numbers is to the specified number systems.

   (i) $(AB7)_{16} = (?)_{10}$

   (ii) $(2740)_8 = (?)_{10}$

   (iii) $(256)_{10} = (?)_2$ $(3 \times 3 = 9)$

2. (a) What are NOR and NAND gates? Why are they called universal gates? Give truth tables for 3-input NAND and NOR gates. (14)

   (b) Explain XOR operation. Give truth tables for 2 input XOR gate. (6)

3. State and prove the basic axions of Boolean algebra with the help of logic circuits. (20)

4. (a) With a neat diagram explain the working of a full adder. Draw the circuit using NAND gates. (10)

   (b) Implement a full subtracor with two half-subtractors with the addition of one inverter circuit. (10)

5. (a) Explain what you understand by don't care conditions. (6)

   (b) Draw Karnaugh map and simplify the following Boolean functions :

   $Y(A,B,C,D) = \Sigma m$ (0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15). (14)

6. (a) Explain the working of R-S and J-K flip flops using logic gates. (10)

   (b) Sketch the circuit of a shift register and explain its working. (10)

7. Discuss the working principle of a synchronous counter with it block diagram. List out its advantages. (20)

8. (a) Explain in detail about the working principle of P-N diode and its characteristics. (15)

   (b) Give the circuit symbols for n-channel and p-channel JFETs. (5)

DE–5831 **12**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2013.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

All questions carry equal marks.

1. Discuss in detail about the developments in communication technology. (20)

2. Explain in detail about the overview of the revolution in computers and communications. (20)

3. Explain in detail about different kinds of software. (20)

4. Explain about communication networks. (20)

5. (a) Describe the various factors affecting data transmission. (10)

   (b) Write short notes on data storage. (10)

6. Write short notes on :

   (a) Modems (8)

   (b) Communication software. (12)

7. Explain the concepts of data management systems. (20)

8. Write short notes on :

   (a) HTML (10)

   (b) XML. (10)

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DE–5832 **13**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2013.

PRINCIPLES OF OPERATING SYSTEM

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE of the following questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the components of the operating system. (10)

   (b) Explain in detail about the operating system structure. (10)

2. (a) Define process. Explain the process states and their transition with a suitable diagram. (10)

   (b) Explain briefly about Race conditions and critical sections. (10)

3. (a) Explain about deadlock avoidance. (12)

   (b) Explain about I/O devices and device controllers. (8)

4. Explain the various page replacement algorithms.

5. (a) Explain about directories. (10)

   (b) Write about file system reliability. (10)

6. Explain in detail about the various system calls. (20)

7. (a) Explain briefly about monitors and message passing. (10)

   (b) Write about priority scheduling and round robin scheduling. (10)

8. Explain about

   (a) Multiprogramming with fixed partitions. (10)

   (b) Multiprogramming with variable partitions. (10)

DE–5833 **14**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2013.

C PROGRAMMING

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain about procedures and recursion. (10)

   (b) What do you mean by text processing? Explain. (10)

2. (a) Briefly explain the operators available in C. (10)

   (b) Write short notes on the following :

      (i) Constants. (5)

      (ii) Variables. (5)

3. (a) (i) Discuss Reading and writing a character with suitable examples. (5)

      (ii) Explain switch statement with example. (5)

   (b) Write a C program to illustrate the concept of two dimensional arrays. (10)

4. (a) Discuss briefly about category of functions. (10)

   (b) Define pointer. Illustrate the concept of pointers with example program. (10)

5. (a) Define structure. Give an example for structures within structures. (10)

   (b) Explain the different storage classes. (10)

6. (a) Explain any four file operation functions with an example. (10)

   (b) Discuss briefly about Macro substitution and conditional substitution. (10)

7. (a) Write a C program which calls a function reverse( ) which accepts a string and display its reverse. (10)

   (b) Discuss about data types of C. (10)

8. (a) Write a C program to copy the contents of one file to another file. (10)

   (b) Define union. Illustrate the concept of union with example program. (10)

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DE–5834 **15**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2013.

ACCOUNTING FUNDAMENTALS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. (5 × 20 = 100)

1. (a) Explain the functions of accounting. (10)

   (b) Distinguish between cost accounting and management accounting. (10)

2. (a) What are accounting conventions? (10)

   (b) Describe the rules of accounts with suitable example. (10)

3. (a) Explain the procedure for preparation of the final account of sole trader. (10)

   (b) How do you find profit or loss under conversion method? (10)

4. (a) What are the limitations of ratio analysis? (10)

   (b) What are 'Turnover Ratio'? Explain their importance. (10)

5. Journalise the following transactions. (20)

1994

Jan 1 Mr. Kumar commenced business with cash Rs. 50,000, Buildings Rs. 1,00,000, Bank Balance of Rs. 1,00,000.

3 Purchased goods for cash Rs. 7,000.

4 Purchased goods for Rs. 7,500 from Dhanam and Co.

5 Sold goods to swamy and co Rs. 9,500

6 Sold good to Nithya and co Rs. 6,500.

7 Sold goods Rs. 2,000

8 Obtained a loan from Indian Bank Rs. 20,000

9 Purchased Furniture Rs. 2,000

10 Purchased Land by issuing a cheque Rs. 25,000

6. From the following Trial Balance of Thiru. Rehman as on 31st March 1995, Prepare Trading and Profit and Loss A/c and Balance Sheet taking into account the adjustment: (20)

Debit balance	Rs.	Credit balance	Rs.
Land and Buildings	42,000	Capital	62,000
Machinery	20,000	Sales	98,780
Patents	7,500	Return Outwards	500
Stock 1-4-1994	5,760	Sundry Creditors	6,300
Sundry debtors	14,500	Bills payable	9,000
Purchases	40,675		
Cash in hand	540		
Cash at Bank	2,630		
Return Inwards	680		
Wages	8,480		
Fuel and power	4,730		

Carriage on sales	3,200	
Carriage on purchase	2,040	
Salaries	15,000	
General expenses	3,000	
Insurance	600	
Drawings	5,245	
	1,76,580	1,76,580

Adjustment:

(a) Stock on 31-3-1995 was Rs. 6,800.

(b) Salary outstanding Rs. 1,500.

(c) Insurance Prepaid Rs. 150.

(d) Depreciate machinery @ 10% and patents @ 20%.

(e) Create a provision of 2% on debtors for bad debts.

7. From the following details, prepare trading and profit and loss account and balance sheet: (20)

	As on 1-1-95	As on 31-12-95
Creditors	37,500	43,750
Furniture	2,500	2,500
Cash	6,250	10,000
Debtors	62,500	87,500
Stock	25,000	12,500

Other Details:

Drawings Rs. 10,000; Bad debts Rs. 1,250; Discount received Rs. 3,750; Discount allowed Rs. 2,500; Sundry expenses Rs. 7,500; Payment to creditors Rs. 1,12,500; Collection from debtors Rs. 1,33,750; Sales returns Rs. 3,750; Purchase returns Rs. 1,250; Charges 5% depreciation on furniture.

8. From the following information make out a balance sheet with as many details as possible: (20)

(a)	Gross profit turnover ratio	25%
(b)	Debtors velocity	3 months
(c)	Creditors velocity	2 months
(d)	Stock velocity	8 months
(e)	Capital turnover ratio	2.5 times
(f)	Fixed assets turnover ratio	8 times

Gross profit for the year ended 31st December 1992 was Rs. 80,000. There was no long-term loan or overdraft. Reserves and Surplus amounted to Rs. 28,000. Liquid assets were Rs. 97,333. Closing stock of the year was Rs. 2,000 more than the opening stock. Bills receivable and bills payable were Rs. 5,000 and Rs. 2,000 respectively.

DE–5835 **21**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

MICROPROCESSORS AND PERSONAL COMPUTERS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What is bus? Explain why data bus is bidirectional and address bus is unidirectional. (10)

   (b) Explain the evolution of microprocessors in detail. (10)

2. (a) What are the operations performed by ALU of 8085? Explain. (5)

   (b) What is flag? List the flags of 8085. (5)

   (c) Explain the general purpose registers in 8085 microprocessor. (10)

3. (a) What is machine cycle? List the various machine cycles in 8085. (10)

   (b) Explain the various addressing modes in 8085. (10)

4. (a) Write an assembly language program to subtract two numbers of 8 bit data stored in memory 4200_H and 4201_H. (10)

   (b) Write an assembly language program to divide two numbers of 8 bit data stored in memory locations 4204_H and 4205_H. Store the remainder in 4202_H and the quotient in 4203_H. (10)

5. (a) What is RAM? Compare the static RAM and DRAM. (10)

   (b) How the ROM memories are classified? Explain. (10)

6. (a) Explain the Isolated I/O and memory mapped I/O methods. (10)

   (b) What is programmed I/O? Explain. (10)

7. (a) Describe in detail about the motherboard and I/O Cards. (10)

   (b) Give a brief note about the ISA, VESA and PCI. (10)

8. (a) Explain the passive preventive maintenance procedures in detail. (10)

   (b) Explain about the Data back up system. (10)

DE–5836 **22**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

OFFICE AUTOMATION

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is tool bar? List and explain any two types of tool bars in windows. (10)

   (b) What is a document? How will you open, save and naming the document? (10)

2. (a) Describe about font dialog box in word. (10)

   (b) Explain about inserting moving and resizing pictures. (10)

3. (a) Explain how to add and remove rows and columns with an example. (10)

   (b) Write the steps for creating students mark sheet and calculate the total and average. (10)

4. (a) How can you change the view and move fields in access? Give an example. (10)

   (b) Discuss about report wizard in Access. (10)

5. (a) Give short note on slide transition. (10)

   (b) Explain about creating and printing binders in detail. (10)

6. (a) How do dialog boxes help us? Explain with an example. (10)

   (b) Discuss formatting toolbar in word. (10)

7. (a) What are the functions used in excel? Explain in detail. (10)

   (b) How can you add, edit and delete fields in Access? Explain with an example. (10)

8. (a) How will you create forms? Explain autoform in access. (10)

   (b) Write a short note on custom animation using in power point presentation. (10)

DE–5837 **23**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

SYSTEMS ANALYSIS AND DESIGN

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define system and discuss its concepts. (5)

   (b) Briefly discuss on DSS, TPS and OAS. (9)

   (c) How abstract system differs from physical system? (6)

2. (a) Explain the technique for conducting the test for Project Feasibility. (5)

   (b) Mention the steps involved in conducting the investigation. (5)

   (c) Discuss the following fact finding techniques (i) interview (ii) questionnaire. (10)

3. (a) How to analyse systems data? (4)

   (b) Discuss on the operational phase in feasibility study. (6)

   (c) How Break Even is occured in cost benefit analysis? (10)

4. (a) Give the concept of drawing data flow diagram. (4)

   (b) Draw the data flow diagram for the transactions that occur in library management system. (6)

   (c) Explain the usage of (i) decision table (ii) decision tree. (10)

5. (a) How logical system design differs from physical system design? (5)

(b) Explain the technique of capturing data for input. (7)

(c) How to perform the input design for on-line systems? (8)

6. (a) Define a file. Mention its types. (5)

(b) Discuss the concept of database. (5)

(c) Explain the following databases

(i) relational

(ii) hierarchical. (10)

7. (a) Compare and contrast system flow chart with program flow chart. (5)

(b) Discuss the following :

(i) HIPO

(ii) VTOC. (10)

(c) List out the importance of system testing. (5)

8. (a) What is the importance of training? Discuss its methods. (8)

(b) How to perform the site preparation? (4)

(c) Give the importance of system maintenance. (8)

DE–5838 **24**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

OBJECT ORIENTED PROGRAMMING AND C++

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Discuss the basic principles in OOP with example. (10)

   (b) Explain the data types in C++. (10)

2. (a) Define encapsulation. Explain data encapsulation with sample programs. (10)

   (b) Explain data encapsulation and friend function. (10)

3. (a) Explain the following :

   (i) This pointer and (5)

   (ii) New and delete operators. (5)

   (b) Discuss the problems with pointer reference. (10)

4. (a) Explain about the polymorphism. (10)

   (b) Discuss the advantages and disadvantages of polymorphism. (10)

5. (a) Explain about the inheritance. (10)

   (b) Discuss about the pure virtual function. (10)

6. (a) Write a sample program for base class and derived class. (10)

   (b) Write a program for virtual base class and abstract class. (10)

7. (a) Explain the unary operators in overloading. (10)

   (b) Describe overloading binary operators in detail. (10)

8. (a) Explain about dynamic constructor. (10)

   (b) Explain about the function overloading. (10)

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DE–5839 **25**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

COMPUTER GRAPHICS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Enumerate the various applications of computer graphics. (10)

   (b) Give a detailed note on liquid crystal display (LCD). (10)

2. (a) Explain the incremental algorithm to draw a straight line. (10)

   (b) Discuss in detail

      (i) Flood filling

      (ii) Boundary filling. (10)

3. (a) How to represent 2D object in a matrix form? Explain. (10)

   (b) Discuss the following basic transformations :

      (i) Translation

      (ii) Rotation

      (iii) Scaling. (10)

4. (a) Write a detailed note on windows and viewports. (10)

(b) Explain the procedure for line segment clipping. (10)

5. (a) Expatiate coordinate transformation. (10)

(b) With an example, explain the 3D composite transformation. (10)

6. (a) Describe the three-dimensional clipping algorithm. (10)

(b) What is hidden surface? Explain it with an algorithm. (10)

7. (a) Enumerate the various components of the user interface. (10)

(b) List out the key issues that are involved in the design of the command language. (10)

8. (a) Describe the various components of the flat bed plotter. (10)

(b) Write the general form of a scaling matrix with respect to a fixed point $p(h, k)$. (10)

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DE–5840 **31**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

BUSINESS COMMUNICATION

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Bring out the importance of business communication. (10)

   (b) What are the various obstacles to communication. (10)

2. (a) Write the special features of a business letter. (10)

   (b) Describe the different parts of a letter with an example. (10)

3. (a) Outline the steps to be taken while preparing a report. (10)

   (b) Draft a report on the need to introduce some incentive schemes to boost the sales of the company. (10)

4. Draft the minutes of the annual general meeting of a company. (20)

5. (a) Write some of the factors that can make a speech interesting. (10)

   (b) Write an essay on "Mobile phone A blessing or curse". (10)

6. What are the essential requirements of an application for employment? (20)

7. Write your resume for the application for the post of an accountant. (20)

8. Write a letter to a tea company, complaining about the tea packets sent by them are less than the weight mentioned in the packets. Ask them to compensate the loss. (20)

DE–5841 **32**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

WINDOWS AND VISUAL BASIC

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) What are the hardware and software required for windows? (10)

   (b) Discuss about file menu and edit menu. (10)

2. (a) Explain about paintbrush. (10)

   (b) What is PIF editor? Explain in detail. (10)

3. (a) Write the important properties and methods of forms. (10)

   (b) Give short notes on Input box. (10)

4. (a) Explain in detail about project explorer window and properties window. (10)

   (b) Write short note on :

      (i) For.....Next

      (ii) Do loops. (10)

5. (a) What is recursive function? Explain with an example program. (10)

   (b) What is control array? How to add and remove control arrays at run time? (10)

6. (a) Write a program for creating fibonacci series. (10)

   (b) Describe picture attributes. (10)

7. (a) What are the types of combo box? List some properties of combo box. (10)

   (b) Explain code window in VB. (10)

8. (a) Write short notes on :

   (i) Frames

   (ii) Check box. (10)

   (b) Define MDI. Write a VB program to illustrate MDI forms. (10)

DE–5842 **33**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

INTERNET AND JAVA PROGRAMMING

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. With a neat figure explain the functions of the various layers of OSI/ISO network model. (20)

2. Write notes on :

   (a) Types of Network topologies. (10)

   (b) Client/Server Architectur. (10)

3. (a) Discuss about E-mail. (10)

   (b) Explain about Functionality of the resolver. (10)

4. (a) Write notes on HTTP Client Request structure. (10)

   (b) Discuss about URI and URL. (10)

5. (a) Explain the usage of Internet in Education. (10)

   (b) Briefly explain about Entertainment on Internet. (10)

6. (a) Explain in detail about data types in Java. (10)

   (b) Explain the concept of Array in Java with example. (10)

7. (a) Discuss about Interface in Java. Write a Java program to illustrate the use of interface. (10)

   (b) Explain the String handling functions in Java with examples. (10)

8. (a) Discuss Input Stream and Output Stream with an example. (10)

   (b) Explain Constructor with example. (10)

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DE–5843 **34**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

RDBMS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

(5 × 20 = 100)

1. (a) Write notes on Database Schema. (10)

   (b) Explain in brief about the guidelines for using Procedure-Oriented Language and Non Procedure-Oriented Language (10)

2. Explain in detail about the Mapping operations of Relational Model. (20)

3. (a) Write notes on Data Definition Facilities of a Relational Model. (10)

   (b) Briefly explain the Data Control facilities of a Relational Model. (10)

4. (a) Explain the Properties of Relational Database. (10)

   (b) Briefly explain the benefits of Oracle. (10)

5. Write SQL queries to

   (a) Create a Table Called "STUDENT" with the fields like Reg.No., Name, Sex, Total, Grade, Result (7)

   (b) Display all the Students detail. (4)

   (c) Display all the students records whose result is fail. (4)

   (d) Display all the records for total marks between 60 and 80 and the student should be female. (5)

6. (a) Briefly explain the Data Access SQL Commands. (10)

   (b) What are DML commands? Explain with examples. (10)

7. Create the sales table and area table with appropriate fields. Join these tables. Select distinct values from that table. (20)

8. Explain the following with example

   (a) Create Table statement (4)

   (b) Modify table (4)

   (c) Rename table (4)

   (d) Copying table to another table. (4)

   (e) Droping table. (4)

DE–5844 **35**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

MANAGEMENT PRINCIPLES AND TECHNIQUES

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Use artificial variable techniques to solve

Maximize $z = 3x_1 - x_2$

Subject to the constraints : $2x_1 + x_2 \geq 2$

$$x_1 + 3x_2 \leq 3$$

$$x_2 \leq 4$$

$$x_1, x_2 \geq 0.$$

2. A company is faced with a problem of assigning 6 different machines to 5 different operators. The costs are estimated as follows (in 100's of Rs.)

Jobs

		a	b	c	d	e
	1	2.5	5	1	5	1
	2	2	5	1.5	7	3
Machines	3	3	6.5	2	8	3
	4	3.5	7	2	9	4.5
	5	4	7	3	9	6
	6	6	9	5	10	6

Solve the above problem.

3. Consider the project of building a house. The details are given below :

Activity :	A	B	C	D	E	F	G	H	I	J	K
Preceeding activity :	–	A	–	B,C	C	G,H	D	B	F	G	E, I, J
to :	3	3	2	6	4	8	3	2	2	1	19
tm :	5	6	4	10	8	12	7	6	5	2	30
tp :	13	9	6	20	18	22	17	10	14	3	65

Draw the network and calculate the expected duration and its variance.

4. The following table gives the activities in a construction project and other relevant information :

Activity :	1–2	1–3	2–3	2–4	3–4	4–5
Normal time :	20	25	10	12	5	10
Crash time :	17	25	8	6	2	5
Normal cost :	600	200	300	400	300	300
Crash cost :	720	200	440	700	420	600

(a) Draw the network and find the critical path

(b) Crash the activity step by step until the shortest duration is reached.

5. A computer contains 10000 resistors. When any resistor fails, it is replaced. The cost of replacing a resistor individually is Rs. 1 only. If all the resistors are replaced at the same time, the cost per resistor would be reduced to 35 paise. The percentage of surviving $s(t)$ at the end of month t is given below :

t :	0	1	2	3	4	5	6
$s(t)$:	100	97	90	70	30	15	0

What is the optimum replacement plan?

6. A machine costs Rs. 10,000. Operating costs are Rs. 500 per year for the first five years. In the sixth and succeeding years operating costs increases by Rs. 100 per year. Assuming a 10% discount rate of money per year, find the optimum length of time to hold the machine.

7. Define Management. Discuss the various functions of management.

8. What do you mean by organization? Explain the principles of organization.

DE–6607 **16**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, MAY 2013.

LAB I — C PROGRAMMING

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

1. (a) Write a C program to print the all ASCII characters.

   (b) Write a C program to create a text file and copy the contents of that file to another file.

---------- Cut here ----------

2. (a) Write a C program to arrange the given set of numbers in descending order.

   (b) Write a C program to create and read the records in an address file.

---------- Cut here ----------

3. (a) Write a C program to accept two strings. Assign a string to another variable, concatenate the two strings and find the length of concatenated string.

   (b) Write a C program to arrange the names in alphabetical order and assign roll number to each name.

4. (a) Write a C program to find the factorial of a given number using function declaration.

   (b) Write a C program to create a text file and display the contents of file.

---------- Cut here ----------

5. (a) A man is paid at the hourly rate of Rs. 50 per hour for the first 50 hours worked. Thereafter, overtime is paid at 2 times the hourly rate for the next 25 hours and 3 times the hourly rate for further hours worked per week. Calculate and print his gross weekly wage.

   (b) Write a C program to find the product of two matrices.

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6. (a) There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes more than 40 marks in each subject then he is declared pass, otherwise fail. Write a C program to do the above.

   (b) Write a C program to accept and display your address using structure.

DE–6608 **26**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

LAB II — C++ AND MS OFFICE

(2003 onwards)

Time : Three hours Maximum : 100 marks

(a) Examiner has to select ONE question for each candidate by Lot system.

(b) Each subdivision carry 50 marks.

(c) Write an algorithm or flow chart for C++.

(d) Write the procedure and draw screenshot for MS Office.

1. (a) Write a C++ program to read student's name and his test score which convrts a test score into its equivalent grade :

(i) Using else if

(ii) Using switch

based on the following conditions :

> = 90 : A

> = 80 and < 90 : B

> = 70 and < 80 : C

> = 50 and < 70 : D

< 50 : E

The output should contain the student's name, test score and his grade.

(b) Create sales table with the following fields :

Name, sex, salescode and salesamount

(i) Add 5 records using forms.

(ii) Display the records for salesamount < 10000

(iii) Display the records for salesamount > 5000 and sex = 'M'.

(iv) Sort the table in the ascending order of salesamount.

2. (a) Write a C++ program to find the product of two matrices.

(b) Create the worksheet with the following fields :

SNo., Reg. No, Name. Age, Tamil, English, Maths, Total, Average, Result.

(i) Compute the total and average column.

(ii) Replace the result with 'PASS' for average > = 40, else 'FAIL'.

(iii) Sort first by name their age.

(iv) Format average column with 2 decimals.

---------- Cut here ----------

3. (a) Write a C++ program to convert given temperature in celsius/Fahrenheit to Fahrenheit/Celsius with option.

(b) Using the MS-Word type a circular letter to be posted to PGDCA candidates to attend PCP. Format the letter and apply the suitable font type and font size.

---------- Cut here ----------

4. (a) Write a C++ program to average the given set of numbers in ascending/descending order.

(b) Create address table with the following fields :

Name, age, sex, street, city and pin.

(i) Add 5 records.

(ii) Display the fields name, street alone on the screen.

(iii) Display the records for age > 20 and city = 'MADRAS'.

(iv) Sort the table in the descending order of name.

5. (a) Write a C++ program to find the square of a given number with different arguments using function overloading.

(b) Create a work sheet with the following fields :

Year, sales and expense.

(i) Draw the graph and mention the appropriate headings.

(ii) Experiment with other types of graph by changing your selections.

(iii) Include more columns (say profit) with data values and create different types of graphs.

---------- Cut here ----------

6. (a) Write a C++ program to create a class objects e1 and e2. The contents of object e1 is added to the object e2 using the operator overloading technique.

(b) Using MS-Office type a paragraph in newspaper format. Also use the following formatting features :

(i) Chose font type wherever needed.

(ii) Use colour options.

(iii) Choose appropriate font size.

---------- Cut here ----------

7. (a) Write a C++ program to read and display the students particulars using class and objects.

(b) Create STUDENT table with the following fields :

Reg. No., Name, Age, Sex, Degree and Percentage.

(i) Add 5 records using forms.

(ii) Display the records for sex = 'F' and percentage between 80 and 90.

(iii) Display the records for degree = 'BSc' and name like 'M*'.

(iv) Sort the table in the descending order of percentage.

8. (a) Write a C++ program to read the derived class data members such as name, null number, sex, height and weight from the keyboard and display the contents of the class on the screen (single inheritance concept).

(b) Create the worksheet with the following fields :

Reg No., Name, Tamil, English, Science, Total.

(i) Fill the total column

(ii) Fill the candidates whose RegNo > = 124

(iii) Filter the candidate whose name = 'RAM'.

(iv) Filter the candidate whose science > 60.

DE–6609 **36**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

LAB III – JAVA AND VISUAL BASIC

(2003 onwards)

Time : Three hours Maximum : 100 marks

Each sub division carries 50 marks.

Examiner has to select questions paper for every student by lot system.

Program writing : 30
Execution : 10
Results : 10
Total : 50

1. (a) Write a VB program to compute simple or compound interest using option buttons, given the values of P, N, R.

   (b) Write a Java program to compute compound interest given the values of P, N, R.

---------- Cut here ----------

2. (a) Write a VB program to compute Gross pay and Net pay of an employee, given the values of basic pay and deductions. Assume your own formulae for Gross pay and Net pay calculations.

   (b) Write a Java program to find biggest and smallest number in an array of 10 numbers.

3. (a) Write a VB program to display odd and even numbers separately between 1 and n, where the value of n is input in a text box.

   (b) Write a Java program to sort 10 characters in an array, in descending order.

---------- Cut here ----------

4. (a) Write VB program to compute electricity Bill for a consumer whose Name, consumer number and units consumed are input in text boxes. Assume your own slabs and rates.

   (b) Write a Java program to compute area and circumference of a circle whose radius is input.

---------- Cut here ----------

5. (a) Write a VB program to display multiplication tables 1 to 5 with 10 entries in each.

   (b) Write a Java program to display the characters in your name vertically (in one per line).

DE–6610 **36**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

LAB III — ORACLE AND VISUAL BASIC

(Common for all branches)

(2003 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by Lot system.

1. (a) Write and test a VB program to print twenty address labels in a two column format for the address :

The Registrar,

University of Madras

Chennai - 600 006.

(b) Create a table BLOOD with the following fields :

DNO, DNAME, DOB, DAGE, DADD1, DADD2, DCITY, DPIN, DBLOOD, DSEX.

(i) Add 5 records

(ii) Display all records

(iii) Display the information of donors in the age group 20 to 25 using BETWEEN option.

2. (a) Write a VB program to compute the surface area and volume of a sphere given the radius. Use option buttons and INPUT Boxes.

(b) Create a table MARKS with the following fields :

REGNO, NAME, M1, M2, M3, TOTAL, AVG, GRADE.

(i) Add 6 records

(ii) Replace TOTAL with M1+M2+M3

(iii) Replace AVG with TOTAL/3

(iv) Replace GRADE with :

A if AVG $\geq$ 60

B if AVG is 40 to 59

C if AVG < 40

(v) Display the maximum marks in each subject.

---------- cut here ----------

3. (a) Write a VB program that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

(b) Create a table LABEL with the following fields :

NAME, STREET, CITY PIN.

(i) Add 6 records and display the contents of the table.

(ii) Display the ascending order sorted list with NAME as primary key.

(iii) Display the contents with appropriate headings.

4. (a) Write and test a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using Label, Text and Option buttons.

(b) Create table BIODATA with the fields :

NAME, AGE, SEX, DEGREE, CITY

(i) Add 5 records and display all male employees.

(ii) Delete all MCA holders and display.

(iii) Display all records having age < 20 and city in Trichy.

---------- cut here ----------

5. (a) Write a VB program to display the day of the given Date. Also convert the given word from lower case to upper case.

(b) Create a table ADDRESS with the following fields :

NAME, STREET, CITY, PIN

(i) Add 5 records and display the structure of a table.

(ii) Add the field DIST

(iii) Fill all DIST with TRICHY

(iv) Change the PIN to 630003 where city is Karaikudi and PIN is 623003.

---------- cut here ----------

6. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely Input, Calculate, Display.

(b) Create a table RESULT with the following fields :

REGNO, NAME, M1, M2, M3, TOT, RES.

(i) Add 5 records

(ii) Fill TOT field with M1+M2+M3

(iii) Fill RES field with Pass if TOT ≥ 150 otherwise FAIL.